GNSS Vulnerabilities and Modern Attacks

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1 Introduction [1]

The term GNSS refers to the set of satellite navigation systems that provide global positioning services and their related augmentation methods. Some examples of satellite navigation systems are the United States' Global Positioning System (first satellite launched in 1978), the Russian GLObal Navigation Satellite System (first satellites launched in 1982), the European Galileo system, and China's BeiDou Navigation Satellite System. GNSS utilizes Medium Earth Orbit (MEO) satellites, located approximately 20,000 km above the Earth's surface, distributed across various orbital planes. The concept is that each constellation consists of a sufficient number of satellites so that at least 4 of them are visible from any point on the Earth's surface: typically, each constellation is composed of 24 or more satellites. The satellites transmit positioning signals that allow users to evaluate their position and velocity with sufficient accuracy. The distance's measure is based on the one-way Time of Arrival (TOA), whereas the velocity's measure is based on the Doppler shift. It is necessary for the satellites within a constellation to be synchronized. However, the receiver can either be synchronized with the satellites, reducing the number of satellites from which it must acquire information to 3, or be unsynchronized, requiring information from 4 satellites to reconstruct its position (it must determine its latitude, longitude, altitude, and its time offset relative to the inertial system time). Every satellite navigation system consists of three segments: space, control, and user. Augmentation techniques act on the space and control segments to improve the accuracy and availability of the positioning service. Furthermore, there is a need to provide navigation information between GNSS receiver update periods and during periods of signal loss or high interference, which is why GNSS systems are usually coupled with other sensors, such as inertial ones. Later, an attack on these systems will be described, which exploits precisely the coupling between INS and GNSS and the operation of the Kalman filter.

1.1 Three-Dimensional TOA Operation

As mentioned previously, the user measures the time it took for the satellite signal to travel the distance separating them, multiplies it by the speed of light, and obtains the satellite-to-user range. Knowing the distance from a single satellite, the user understands that they are located at a point on the sphere centered on the satellite, with a radius equal to that distance. By solving a system of equations for three or more such spheres, two possible locations are identified: in the case of a user located near or on the Earth's surface, it is easy to disambiguate the solutions by discarding the one located at unrealistic altitudes in space; otherwise, the solution selection process is not as trivial and requires the use of auxiliary information, such as prior knowledge of orbital dynamics.

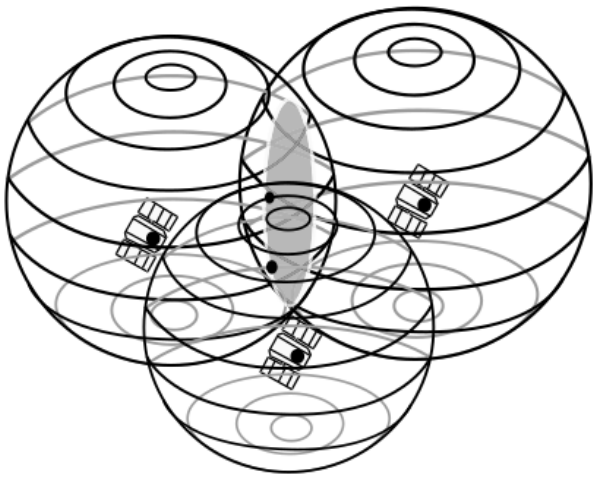


Figure 1: TOA operation example

1.2 GNSS Signal

Most GNSS use carrier signals in the L-band, meaning frequencies from 1 to 2 GHz. This choice is dictated by the fact that lower frequencies are subject to greater delays due to the Earth's atmosphere and signal dispersion caused by the Earth's atmosphere inhomogeneities, while higher frequencies require more transmission power from the satellites and are more heavily impacted by rainfall attenuation. Within the L-band, two sub-bands have been allocated for satellite radionavigation services, specifically 1164–1300 MHz and 1559–1610 MHz. Typically, GNSS use Binary Phase Shift Keying (BPSK) modulation, a digital modulation with two possible phase values (0° or 180°).

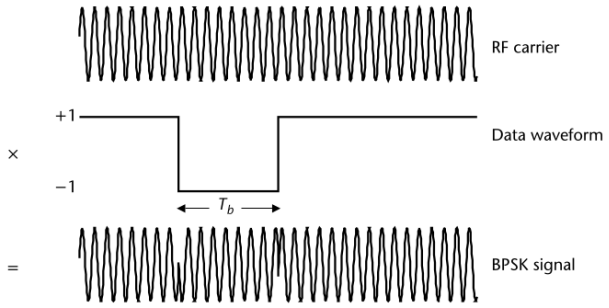


Figure 2: BPSK modulation

Additionally, most GNSS utilize a technique called Direct Sequence Spread Spectrum (DSSS), which consists of modulating the carrier signal with a spreading waveform or Pseudo-Random Noise (PRN). The latter is a deterministic waveform, known to the receivers, characterized by a symbol rate much higher than that of the data waveform. Civil-use GNSS signals employ unencrypted, periodic ranging codes varying in length from 511 to 767,250 chips. Since the bandwidth of a digitally modulated signal is directly proportional to the rate at which its symbols change, the signal modulated with ranging codes will have a wider bandwidth and is referred to as spread spectrum. The use of DSSS allows for:

1. More precise ranging by the receiver, due to the higher frequency of symbol inversions;
2. Code Division Multiple Access (CDMA), meaning the receiver can distinguish signals from different satellites based on their ranging codes;
3. Ensuring a significant rejection of narrowband interference.

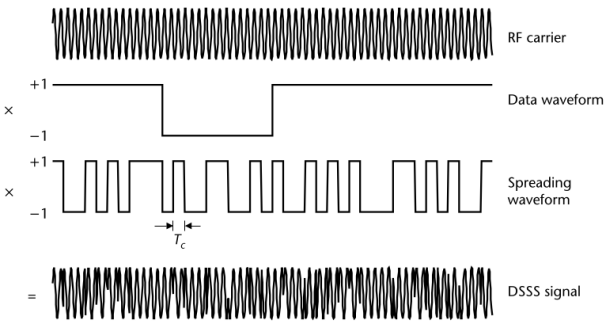


Figure 3: BPSK modulation

2 GNSS Attacks [2]

The underlying problem of GNSS vulnerabilities can be found in the low signal power that actually reaches the Earth's surface. This power is around -160 dBW [3], a value below the thermal noise floor (about -144 dBW for L1 civil signal); the signal is easily subject to interference or obscuration. This

opens up possible attacks on the security of the information provided by satellite navigation services, involving the aspects of confidentiality, integrity, and availability of such information. Furthermore, it is important to note that the development of civilian GNSS technology has not historically included security by design, meaning the presence of mechanisms in the receiver design to ensure the authenticity of the data source or to increase robustness against possible attacks. The latest technologies are attempting to address this critical issue; for example, Galileo is introducing navigation data authentication services, such as OSNMA (Open Service Navigation Message Authentication). The following paragraphs introduce the most common attacks on GNSS.

2.1 Jamming

Since GNSS use RF signals, they are subject to RF interference. Jamming exploits interference to degrade service accuracy or make receiver tracking impossible (DoS attack). The signals typically used by jammers are:

1. *Continuous Wave (CW)*: a pure, unmodulated sinusoid at a specific frequency; minimal impact on GNSS due to the despreading effect of the DSSS modulation and the use of notch filters.
2. *Frequency Modulated Continuous Wave (FMCW) or Chirp jammer*: a sinusoid whose frequency is not fixed but changes continuously over time, sweeping back and forth across an entire frequency range; it largely negates the DSSS processing gain, raising the noise floor across the entire antenna and making satellite tracking impossible if the emitted power is sufficient.
3. *Band-Limited Additive White Gaussian Noise*: thermal noise modulated through a band-pass filter with the same bandwidth as the targeted signal.
4. *Pulsed jamming*: very high power and very short duration pulses; it blinds the receiver antenna's Low Noise Amplifier (LNA), forcing it into its saturation region; during the recovery time, it remains deaf to incoming external signals.
5. *Multi-tone CW*: the jammer transmits multiple pure sinusoids at slightly different frequencies, evenly spaced within the GNSS band; it allows exhausting the number of adaptive notch filters available in the receiver.

2.2 Meaconing

Meaconing consists of the delayed retransmission of GNSS signals. The number of spoofed signals corresponds to the number of satellites in view. The counterfeit signals are identical to the original signals, except for a constant amplification factor and a constant delay. A noise contribution is also added, due to the meaconer's electronics. Consequently, the target receiver produces wrong PVT solutions.

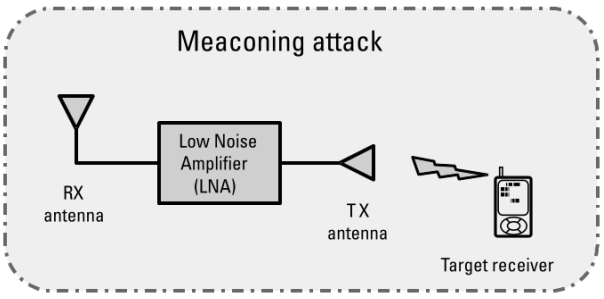


Figure 4: Meaconing attack scheme

2.3 Spoofing

In spoofing, counterfeit signals similar to GNSS ones are transmitted with the intent of causing the victim to generate a false position. Typically, spoofing attacks are classified based on their complexity:

1. *Simplistic attacks*: The attacker generates GNSS signals that are usually unsynchronized with the real signals and inconsistent (for example, they may contain outdated orbital data or data from satellites not visible to the victim). This, combined with the fact that counterfeit signals generally have higher power, makes these attacks easily blockable through appropriate countermeasures. It is necessary for the victim to lose lock on the original signal; this can be achieved by directly manipulating the receiver's cable or antenna, or by jamming the receiver to obscure the original signal.

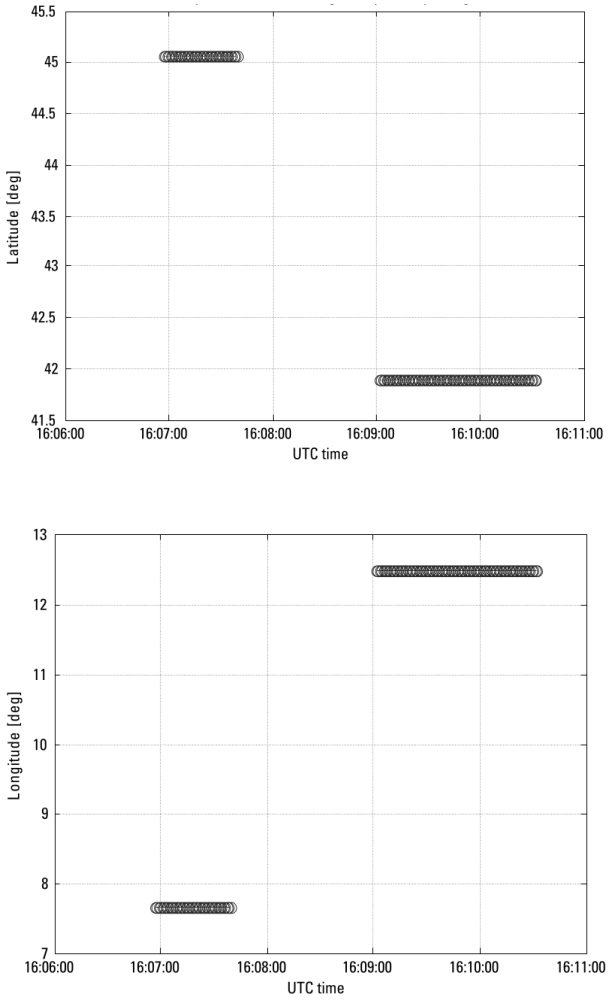


Figure 5: Discontinuity in the Position during a Simplistic Spoofing Attack

2. *Intermediate attacks*: The attacker extracts time, position, and satellite information from the original GNSS signal, then exploits the synchronization of codes and carriers to generate counterfeit signals. Furthermore, the spoofer must know its position relative to the victim and their dynamics in order to generate a plausible signal. Once the code phase alignment is achieved, the spoofer proceeds with the signal lift-off to take control of the receiver.

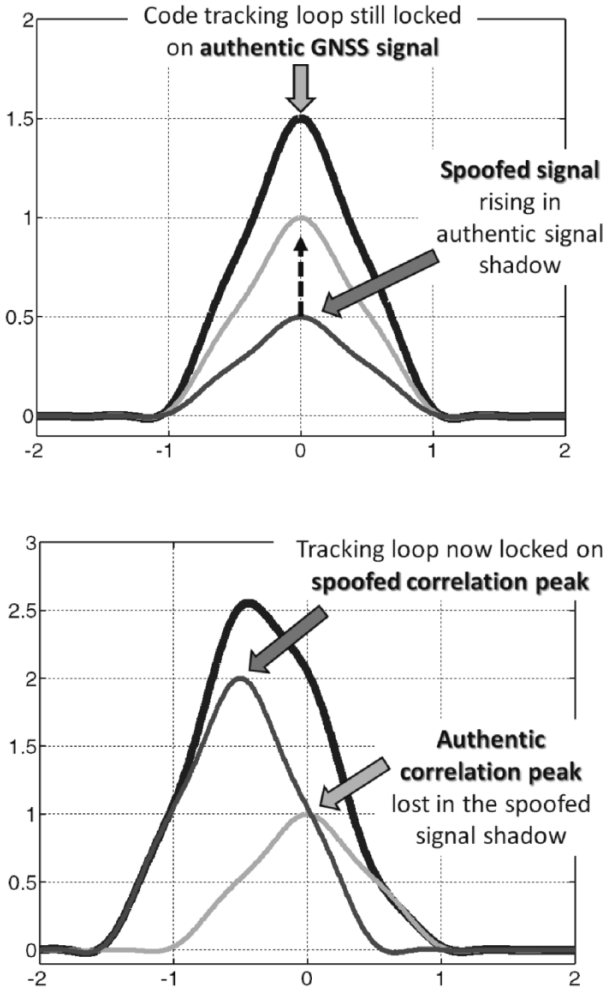


Figure 6: Illustration of an intermediate spoofing attack on a single channel of a GNSS receiver

3. *Sophisticated attacks*: The attacker uses multiple coordinated and synchronized receivers. They must also know the phase center of both the victim's and their own antennas in order to generate, by exploiting the constructive properties of the signals, a GNSS signal with a tilted synthetic wavefront, so as to simulate the true orbital position of the satellite. Such spoofers are also capable of suppressing authentic signals at the target receiver's antenna. These attacks evade complex countermeasures, such as those based on the Angle of Arrival (AoA). However, constructing such sophisticated spoofers is difficult, lowering the probability of these attacks.

2.4 Hybrid/Combined Attacks

2.4.1 Relaying attacks

A modified version of the classic meaconing attack. The idea is to transport the recorded signal kilometers away and then retransmit it, attacking nearby receivers. To implement this attack, the attacker's receiving and transmitting antennas are placed at a significant distance, which implies that they cannot communicate via a direct analog link (this would require high transmission power and signal amplification along the path, and the signal would suffer such attenuation that it would disappear during the journey). A high-speed radio link is therefore established between the two components: the recorded analog signal is converted into a sequence of bits by an Analog-to-Digital Converter, a receiver-side modem compresses, encapsulates, and sends the data over the channel, a transmitter-side modem receives the bitstream and channels it into a Digital-to-Analog Converter, and the resulting signal is amplified and radiated through an antenna towards the victim. The relaying attack is one of the most dangerous because the attacker

does not modify or forge the signals, but simply samples the signal and reconstructs it elsewhere, preserving the signal's cryptographic authenticity. It should be noted that the retransmitted signal may be inconsistent given the considerable distance between the receiving and transmitting components: the satellites visible to the victim might be different. Countermeasures to this attack involve integrating the GNSS with other sensors to verify the validity of the PVT solutions. The attack is nevertheless complex, especially if the target is not static: the remote antenna must be moved following a plausible trajectory without being noticed.

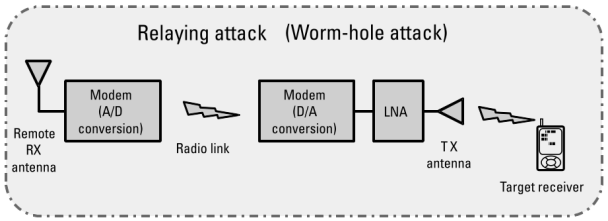


Figure 7: Relaying attack scheme

2.4.2 Meaconing with Variable Delay

The basic idea of this variant is not to have a constant time delay, as happens in classic meaconing, in order to evade countermeasures based on clock drift monitoring. By utilizing high-performance digital signal processing (DSP) hardware platforms, the attacker is able to minimize the introduced delay. As the delay approaches zero, the counterfeit signal and the authentic one become perfectly code phase aligned. This alignment enables a seamless liftoff of the receiver's tracking loops, subsequently allowing the meaconer to gradually increase the delay at a rate consistent with the target receiver's clock drift, until imposing a significant time offset. This type of attack represents a serious threat, especially for applications that rely on the time information provided by GNSS. Variable delay meaconing can be cascaded with other techniques to launch even more sophisticated multi-stage attacks. For example, it can serve as a preliminary phase to stealthily accumulate a delay on all received GNSS signals; once a sufficient time margin is obtained to reliably estimate the navigation data bits, the attacker can trigger a second stage by introducing arbitrary manipulations of the signal, effectively evolving into a SCER attack.

2.4.3 Security Code Estimation and Replay (SCER) Attack

This attack is designed to target GNSS signals protected by cryptographic defenses. The fundamental characteristic of SCER is that it operates entirely at the physical layer. This aspect explains why cryptography fails: while traditional defense systems aim to protect the data layer, the attacker acts directly on the raw radio wave. To implement this attack, it is necessary to employ a high-gain antenna that allows the attacker to perform a chip-by-chip estimation of the incoming signal. In this way, the attacker listens to the original signal and faithfully replicates it, maintaining perfect cryptographic coherence. However, the necessity to estimate the original bits on the fly inevitably introduces an observation delay into the counterfeit signal. The attacker retransmits the signal from each satellite, adding a control delay to the observation delay in order to induce the victim to generate incorrect PVT solutions.

2.4.4 Meaconing or Spoofing Plus High-Gain Antennas

This attack is designed to be effective even against cryptographically protected GNSS signals. High-gain directional antennas are employed, each pointing at a specific satellite (at least 4 antennas are required). The use of such highly directional antennas allows for gathering a signal powerful enough to raise the data bits and the spreading code above the thermal noise floor. In this way, it is possible to directly observe these components without the need to perform despreading. Based on the achieved signal-to-noise ratio (SNR), the attacker can decide to carry out a spoofing attack or a meaconing variant:

- *Spoofing*: if the SNR is sufficiently high, the attacker demodulates the radio frequency signal and physically estimates the sequence of the individual chips. In this way, the attacker regenerates the

satellite signal components to produce a clean and plausible counterfeit signal, to which selective delays are then applied.

- *Modified meaconing*: if the signal is not clean enough to reliably estimate its components, the attacker simply collects the raw wave from each individual antenna and delays it, then mixes each component into a single signal. The weakness of this modified meaconing is twofold: firstly, the noise captured by each antenna is summed into the final signal, significantly increasing the total noise perceived by the victim receiver; secondly, any signals from other satellites not perfectly suppressed by the antenna's directivity will be included in the transmission, providing valuable clues for anti-spoofing countermeasures.

From a practical point of view, however, these attacks are complex and expensive to implement, severely limiting their feasibility, especially in scenarios with many satellites in view or with moving target receivers.

2.5 Attacks Summary

Table 1: Summary of GNSS Attacks and Key Characteristics

Attack Category	**Specific Type**	**Mechanism & Key Characteristics**
Jamming	Classic Jamming	Exploits RF interference to degrade service accuracy or prevent receiver tracking (DoS attack). Techniques include Continuous Wave (CW), FMCW/Chirp (negates DSSS processing gain), Band-Limited Noise, Pulsed jamming (blinds the LNA), and Multi-tone CW (exhausts adaptive notch filters).
Meaconing	Classic Meaconing	Delayed retransmission of authentic GNSS signals. Counterfeit signals are identical to the originals but introduce a constant time delay, a constant amplification factor, and electronic noise.
Spoofing	Simplistic	Generates unsynchronized, inconsistent, high-power counterfeit signals. Requires the victim to lose lock on the authentic signal via physical manipulation or preliminary jamming. Easily blocked by countermeasures.
	Intermediate	Requires knowledge of the target's relative position and dynamics to execute a seamless signal lift-off, align the counterfeit signals, and take control of the receiver.
	Sophisticated	Uses multiple coordinated and synchronized receivers to generate a tilted synthetic wavefront, simulating the true orbital position. Capable of suppressing authentic signals and evading complex defenses, though highly difficult to implement.
Hybrid / Combined	Relaying Attack	Executes long-distance meaconing by converting recorded analog signals into a bitstream, transmitting them via a high-speed digital radio link, and reconstructing them. Preserves cryptographic authenticity but may expose inconsistencies in the visible satellite constellation.

Table 1: Summary of GNSS Attacks and Key Characteristics (continued)

Attack Category	**Specific Type**	**Mechanism & Key Characteristics**
	Variable Delay Meaconing	Evades clock drift monitoring by using high-performance hardware to minimize initial delay, achieving perfect code phase alignment and seamless liftoff. The delay is then gradually increased. Can escalate into multi-stage attacks like SCER.
	SCER (Security Code Estimation and Replay)	Operates strictly at the physical layer to bypass cryptographic data protection. Uses a high-gain antenna for on-the-fly, chip-by-chip signal estimation. Retransmits the signal by adding a control delay to the observation delay to induce false PVT solutions.
	Spoofing / Meaconing + High-Gain Antennas	Employs at least 4 highly directional antennas to raise signals above the thermal noise floor. High SNR enables clean signal regeneration with selective delays (Spoofing); lower SNR leads to mixing raw delayed waves (Modified Meaconing), which significantly increases total perceived noise and exposes clues to countermeasures.

3 Recent Attacks

The escalation of threats to GNSS systems between 2023 and 2025 marked a watershed in global navigation security, transforming what were once considered purely theoretical vulnerabilities into practical incidents. In recent years, attack techniques have largely surpassed simple jamming, evolving into sophisticated manipulations capable of deceiving the most advanced sensor fusion algorithms. In the context of civil aviation, between the end of 2023 and the beginning of 2024, an alarming wave of covert spoofing attacks was recorded along the air corridor between Iraq and Iran. As documented by independent investigations by OPSGROUP [4] and confirmed by safety bulletins from the European Union Aviation Safety Agency (EASA) [5], these attacks employed seamless takeover techniques that were not limited to falsifying GPS coordinates, but induced mathematical poisoning in the Inertial Reference Systems (IRS) of commercial aircraft, causing the cascading collapse of primary navigation instruments. Concurrently, macroscopic-scale spoofing has emerged as a tool for interdiction and electronic defense in urban environments. Between 2023 and 2024, satellite monitoring conducted by the Radionavigation Laboratory at the University of Texas at Austin [6], corroborated by international journalistic investigations [7], certified a massive location spoofing event in the Tel Aviv area. In order to confuse the guidance systems of hostile drones, signals across the entire metropolitan area were deliberately altered, forcing civilian receivers to calculate false positions located at the Beirut airport. The repercussions of modern electronic warfare have also compromised commercial operations in Eastern Europe. In April 2024, severe and persistent anomalies in GNSS signals, traced by Western intelligence to emitters distributed in the Russian exclave of Kaliningrad, forced the airline Finnair to suspend routes to Tartu Airport in Estonia [8], effectively establishing cross-border electronic no-fly zones. Finally, GNSS vulnerabilities have been actively exploited in the maritime domain for purposes of geopolitical concealment. Open-source intelligence dossiers by the Center for Advanced Defense Studies (C4ADS) highlighted how, between 2024 and 2025, numerous commercial vessels transiting the Red Sea and the Strait of Hormuz employed Software-Defined Radios (SDR) to implement local auto-spoofing [9]. This technique allowed the direct manipulation of AIS (Automatic Identification System) transponders, generating ghost trajectories to evade international monitoring and sanctions. In light of this rapid evolution of threats, this article aims to illustrate the architecture of loosely coupled navigation systems (GNSS/INS Loosely Coupled) and their related integrity monitoring mechanisms, and subsequently present an in-depth analysis of how modern covert spoofing techniques

manage to manipulate the trajectories of Unmanned Aerial Vehicles (UAVs) without triggering system alarms.

4 INS/GNSS Systems [1]

As mentioned in the introductory paragraph, the need to provide continuous navigation between GNSS receiver sampling periods and during periods of signal obscuration and high RF interference has led to the integration of GNSS receivers with other sensors. Typically, Inertial Navigation Systems (INS) are employed, and the Kalman filter is used as integration method.

The GNSS and INS systems are structurally complementary: the former provides a positioning service with an upper-bounded error but is vulnerable to electromagnetic interference; the latter is a completely autonomous system, yet its accuracy degrades over time. Specifically, INS systems utilize gyroscopes and accelerometers to evaluate angular velocity and acceleration along the three Cartesian axes, respectively.

Both types of sensors are subject to biases, which impact the growth of the position error calculated by the INS. The following approximations describe this growth in the case of short-term GNSS signal outages (less than 10 minutes):

$$\delta p = b_{acc} \frac{\Delta t^2}{2} \tag{1}$$

$$\delta p = g b_{gyr} \frac{\Delta t^3}{6} \tag{2}$$

where b_{acc} and b_{gyr} represent the uncalibrated biases of the accelerometer and gyroscope, respectively, g is the acceleration due to gravity, and Δt is the duration of the GNSS interruption.

It can be observed that a residual accelerometer bias produces a quadratic growth in position error, while a residual gyroscope bias results in a cubic growth of this error, making external aid indispensable for the continuous calibration of the sensors.

4.1 Architectural Evolution

Initially, GNSS and INS systems were coupled by transmitting their respective outputs to a central navigation processor which, through a Kalman filter, calculated the final PVT solution. This architecture, defined as *loosely coupled*, keeps the two systems autonomous but exposes the system to the so-called *cascaded filter problem*. If the GNSS receiver also employs an internal Kalman filter to calculate its own position, the data sent to the main navigation filter is no longer affected by pure white noise, but rather by an error temporally correlated to the previous state. Consequently, the second filter tends to rely increasingly on GNSS solutions and neglects the real dynamics provided by the INS, leading to a potential divergence of the estimate in the event of sudden maneuvers or attacks.

To overcome this criticality, the modern standard employs a *tightly coupled* architecture in which the two systems are fused at both hardware and logical levels. In this configuration, the GNSS receiver no longer calculates an autonomous positional solution but provides the navigation processor directly with the raw measurements of each visible satellite, namely pseudoranges and Doppler measurements. This approach allows the designer to perform adaptive tuning of the measurement noise covariance matrix relative to individual satellite channels. By evaluating real-time quality metrics, such as the signal-to-noise ratio of the raw data, the algorithm adapts the Kalman Gain. In this way, the filter assigns predominant weight to clean signals and mathematically neutralizes the impact of degraded signals.

4.2 Error-State Kalman Filter

In modern integrated navigation systems the Kalman filter does not directly calculate the absolute position of the vehicle but exclusively estimates its error vector. This formulation is known as the *Error-State Kalman Filter* (ESKF). The system state is modeled through three components: the true state (the actual and unknown physical kinematics of the vehicle), the nominal state (the kinematics calculated by high-frequency integration of the INS data, without considering noise and drifts), and the error state, which is the mathematical discrepancy between the first two.

The Kalman filter produces estimates of the error vector δx at successive discrete time epochs, starting from a vector of noisy GNSS measurements z, characterized by a measurement noise covariance matrix

R. The filter state variables relative to the INS follow the error dynamics, represented by the differential equation:

$$\frac{d(\delta x)}{dt} = F(\delta x) + b \tag{3}$$

where F is a matrix expressing the unforced error dynamics of the INS (the physical model describing how position, velocity, and attitude errors propagate and couple over time), while b represents the intrinsic sensor error (such as gyroscope and accelerometer bias). The total number of filter states depends on the desired level of real-time calibration.

The ESKF algorithmic cycle consists of three main phases:

1. *Prediction (high frequency):* During vehicle motion, the algorithm continuously integrates INS data to propagate the nominal state. Since the filter works in discrete time, the differential equation is approximated by calculating the discrete-time state transition matrix Φ via a Taylor series expansion of the matrix F: $\Phi \approx I + F\Delta t + \frac{F^2\Delta t^2}{2}$. Depending on the severity of the vehicle dynamics, a first or second-order approximation is adequate. The matrix Φ is used to propagate the error covariance matrix (P), which tracks the filter's uncertainty.

   (a) *Nominal State Integration:* The algorithm integrates the kinematic equations of motion to update the nominal state:

   $$\hat{x}_{k|k-1,ins} = A\hat{x}_{k-1,ins} + B\hat{a}_{k-1,ins} \tag{4}$$

   Since no external observations have arrived yet, the expected a priori error for this instant remains theoretically zero ($\delta\hat{x}_k^- = 0$).

   (b) *Error Covariance Propagation:*

   $$P_k^- = \Phi_k P_{k-1}^+ \Phi_k^T + Q_d \tag{5}$$

   where P_k^- is the a priori error covariance matrix, which tracks the filter uncertainty; Φ_k is the state transition matrix; Q_d is the discrete process noise matrix (derived from the continuous noise Q). By being added to P, it represents the increase in uncertainty due to unmodeled dynamics and stochastic sensor drifts.

2. *Correction (low frequency):* Upon the arrival of GNSS data, the filter activates to calculate the accumulated error and correct the system uncertainty.

   (c) *Innovation Calculation:*

   $$\tilde{y}_k = z_k - H_k\delta\hat{x}_k^- \tag{6}$$

   In the ESKF, the measurement provided to the filter (z_k) is the observed error, calculated as the difference between the data provided by the GNSS and the nominal state calculated from the INS data. The innovation $\tilde{y}_k$ therefore represents the difference between this newly measured error (z_k) and the error the filter already expected to encounter due to the INS system drift ($H_k\delta\hat{x}_k^-$). The matrix H_k maps the state space into the measurement space.

   (d) *Kalman Gain Calculation:*

   $$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + R)^{-1} \tag{7}$$

   Through this equation, the ESKF balances trust between the inertial prediction (expressed by the covariance matrix P) and the satellite measurement (expressed by the noise R).

   (e) *A posteriori Error State Estimation:*

   $$\delta\hat{x}_k^+ = K_k\tilde{y}_k \tag{8}$$

   By multiplying the Innovation by the Kalman Gain, the filter obtains the final a posteriori estimate of the error state ($\delta\hat{x}^+$).

(f) *Covariance Update:*

$$P_k^+ = (I - K_k H_k) P_k^- \tag{9}$$

The uncertainty contained in P is reduced accordingly, as the system has benefited from external GNSS information.

3. *Injection and Reset:* In the ESKF, the newly estimated error is not kept in the filter but is actively exploited to heal the drift.

(g) *Error Injection:*

$$\hat{x}_{k,ins}^+ = \hat{x}_{k|k-1,ins} - \delta\hat{x}_k^+ \tag{10}$$

The newly estimated error is injected into the navigation system to correct the nominal state, realigning the calculated trajectory with the real one and recalibrating the sensor biases.

(h) *Error Reset:*

$$\delta\hat{x}_k^+ \leftarrow 0 \tag{11}$$

Following this feedback, the error state vector inside the filter is forcibly zeroed ($\delta x = 0$), making it ready to start accumulating and tracking new drifts in the next cycle.

4.3 ESKF as a Countermeasure to Spoofing and Jamming

A vehicle equipped with an Error-State Kalman Filter can utilize this algorithm as a countermeasure against jamming and spoofing attacks.

In ultra-tight coupled systems the kinematics data computed by the INS are used to pre-compensate and synchronize the receiver's local replica with the satellite signals. This allows for a reduction in the bandwidth at which the tracking loop operates, consequently minimizing the noise and interference power that can penetrate it. This bandwidth restriction, combined with the processing gain provided by the signal despreading mechanism, makes the system inherently more robust to jamming attacks.

Regarding spoofing, a statistical innovation check is implemented. In many designs, the innovation value is compared with its expected variance, defined by the covariance matrix $HPH^T + R$. If the innovation exceeds the statistical thresholds of this variance, the ESKF detects an inconsistency between physical reality (INS) and the radio signal (GNSS). It is therefore necessary to understand the nature of the error to implement appropriate mitigations:

- *GNSS error mitigation:* If the innovation shows sudden anomalies inconsistent with the physical dynamics measured by the sensors, the system identifies a probable spoofing attack. In this case, one or more GNSS measurements are discarded or made less influential in realigning the nominal state. This exclusion prevents the counterfeit signal from poisoning the navigation solution, allowing the system to continue relying on inertial integration and valid satellites only.

- *INS error mitigation:* In situations where discrepancies accumulate and the internally propagated solution (INS) is suspect, the best strategy is often a reset of position and velocity toward the GNSS-only solution. Simultaneously with this reset, the estimates associated with INS errors are zeroed or their error variances (in matrix P) are artificially increased. This forcing compels the ESKF to re-evaluate the estimated biases and recalculate them from scratch based on the future history of GNSS measurements deemed safe.

5 Covert Spoofing on UAVs

Although the Error-State Kalman Filter is applicable as a countermeasure to classic jamming and spoofing attacks, it remains vulnerable to covert spoofing attacks. The idea behind these attacks is to produce counterfeit signals that initially mimic the original satellite signals and subsequently deviate gradually from the actual values to lead the vehicle along a trajectory desired by the attacker. An example of a covert spoofing attack is presented in [11].

In this study, an attack is carried out on the loosely coupled INS/GNSS system used by UAVs. Despite the superior theoretical robustness margins offered by the tightly coupled approach, the vast majority of commercial UAVs on the market still adopt loosely coupled architectures. This engineering

preference is primarily dictated by stringent economic cost constraints, flexibility and modularity in the interchangeability of third-party hardware sensors and reduced computational resource requirements for onboard algorithm processing.

The system is modeled using a 9-dimensional error state vector for the INS component:

$$\delta X = [\delta p, \delta v, \delta a]^T \tag{12}$$

where the three states represent position error, velocity error, and accelerometer bias error, respectively.

A spoofing algorithm is then designed to produce counterfeit signals that do not alert the Error-State Kalman Filter controls, while simultaneously causing the UAVs to deviate toward the desired trajectory. It is important to note that the attacker must be equipped with radar and other assistive detection devices to acquire precise information on the UAV's motion status to achieve precise trajectory control. The algorithm consists of the following sequential phases for each time epoch k:

1. *Target estimated state update:* The spoofer acquires the UAV's real motion state via external radar tracking equipment ($\hat{X}_k^R$) and updates both the current state estimate ($\hat{X}_k^s$) and the uncorrected INS a priori prediction ($\hat{X}_{k|k-1,ins}^s$):

$$\hat{X}_k^s = \hat{X}_k^R \tag{13}$$

$$\hat{X}_{k|k-1,ins}^s = A\hat{X}_{k-1,ins}^s + B\hat{a}_{k-1}^s \tag{14}$$

   where $\hat{X}_{k-1,ins}^s$ is the corrected INS output at the previous step and $\hat{a}_{k-1}^s$ is the estimated UAV acceleration at time $k-1$.

2. *Determination of the spoofer's control acceleration:* The *spoofer* calculates the required acceleration a_k^s to lure the vehicle toward the planned deceptive trajectory $X_k^{sp} = [p_k^{sp}, v_k^{sp}, a_k^{sp}]^T$ using the proportional-derivative matrix K_{PD}, which maps the trajectory tracking error, defined as the difference between the estimated absolute state and the reference trajectory, to the acceleration control quantity:

$$a_k^s = -K_{PD}(\hat{X}_k^s - X_k^{sp}) \tag{15}$$

   Since the UAV responds to its flight controller's commands to follow the original trajectory, the overall fictitious acceleration a^* that the falsified signal must force into the system is equal to:

$$a^* = a_k - a_k^s \tag{16}$$

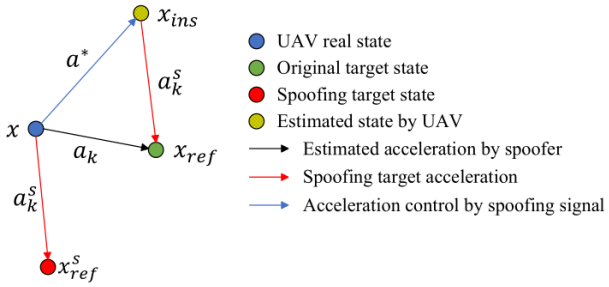


Figure 8: Diagram of acceleration information effect for generating spoofing signals

3. *Calculation of the apparent navigation state:* To ensure the attack remains covert, the spoofer must induce a navigation error in the UAV's flight controller that triggers the required acceleration for the trajectory deviation, while keeping the innovation residuals of the Kalman filter below detection thresholds. Therefore, the estimated state onboard, $\hat{X}_k^s$, is calculated to include the control dynamics (a^*):

$$\hat{p}_k^s = p_{k,ref}^s - \frac{a^* - a_{k,ref}^s}{K_{PD}} \tag{17}$$

$$\hat{v}_k^s = v_{k,ref}^s \tag{18}$$

4. *Calculation and generation of the counterfeit GNSS measurement:* By mathematically inverting the update relation of the UAV's Kalman filter, the spoofer derives the exact falsified position/velocity coordinate ($X^s_{k,gnss}$) that it must induce in the aircraft's satellite receiver:

$$X^s_{k,gnss} = \hat{X}^s_{k|k-1,ins} - (K^s)^{-1}(\hat{X}^s_{k|k-1,ins} - \hat{X}^s_k) \tag{19}$$

where K^s is the steady-state Kalman gain of the drone's filter ($K^s = \lim_{n\to\infty} K_k$).

5. *Update of the spoofer's recursive variables:* Finally, the spoofer updates the estimated UAV acceleration and the corrected INS value for the next epoch:

$$\hat{a}^s_k = \hat{a}^R_k \tag{20}$$

$$\hat{X}^s_{k,ins} = \hat{X}^s_k \tag{21}$$

where $\hat{a}^R_k$ is calculated by differentiating the UAV velocity detected by the radar over a short time interval.

The authors of the paper point out that highly precise instrumentation is required for the attack to succeed. This is due to the fact that, to maintain total invisibility of the deception, the artificially generated radio signals transmitted toward the UAV must not induce a loss of lock in the receiver's tracking loops under any circumstances. Physical signal synchronization constraints dictate that the code phase difference relative to the authentic signal must not exceed 0.1 chip and that the carrier frequency offset must remain strictly below 10 Hz.

The intrinsic errors of the detection instruments directly affect the calculation of the required time delay and the artificial Doppler shift. Through analytical analysis and simulation experiments based on the standard GPS L1 C/A signal, the paper demonstrates that to satisfy the stringent physical tolerances of the hardware and avoid detection or attack failure, the external radar instrumentation must guarantee the following maximum error thresholds:

- *Position detection error* (δp): must be strictly less than 30 meters.
- *Velocity detection error* (δv): must be strictly less than 1 m/s.

If the radar apparatus employed by the attacker exhibits lower accuracy, the introduced time or frequency shift would exceed the tolerance margins of the drone's GNSS receiver discriminator circuits, causing an unexpected interruption in satellite channel tracking or the immediate activation of integrated anti-spoofing alarms.

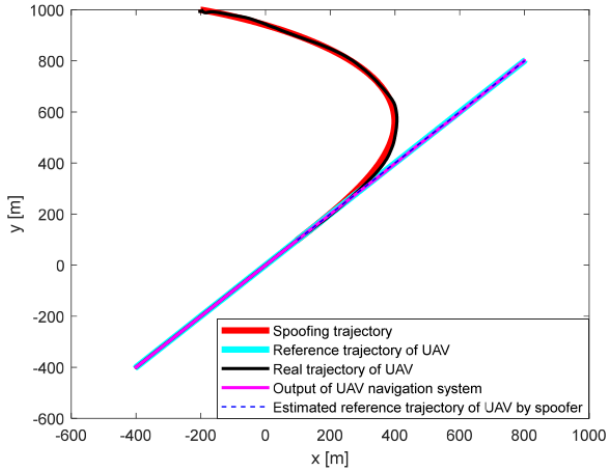


Figure 9: Spoofing effect without Radar detection error

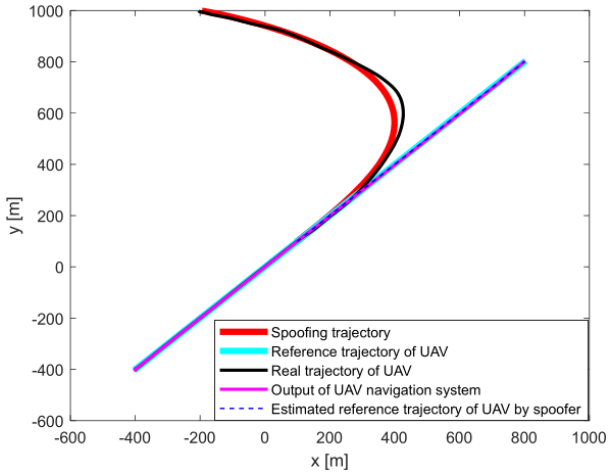


Figure 10: Spoofing effect under high-precision radar detection condition

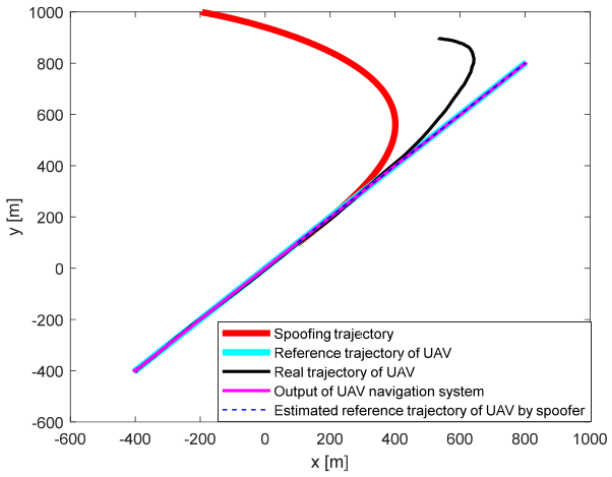


Figure 11: Spoofing effect under low-precision radar detection condition

6 Conclusions

The integration of GNSS receivers with Inertial Navigation Systems (INS) constitutes a robust countermeasure against conventional jamming and spoofing attacks. However, as demonstrated by the examined study, attackers can implement sophisticated counter-countermeasures. By exploiting the equations and internal correction mechanisms of the integration filters (such as the ESKF), a covert spoofing attack is capable of inducing the vehicle to deviate in a controlled manner from its true trajectory, while simultaneously keeping the onboard estimated navigation solution locked onto the original flight plan.

It is therefore necessary to develop and implement advanced detection algorithms to neutralize these threats. One of the currently most recognized strategies relies on the temporal statistical analysis of the Kalman filter innovation vector ($\tilde{y}_k$). Under normal operating conditions, the innovation represents the discrepancy between the new satellite measurement and the inertial prediction. Being dominated solely by the thermal noise of the sensors and unmodeled dynamics, it behaves as a zero-mean stationary stochastic process (white noise), oscillating symmetrically and randomly between positive and negative values around zero. Conversely, during a covert spoofing attack, the attacker injects a fictitious acceleration vector (a^*) to imperceptibly drag the vehicle toward a predetermined route. Even though the malicious algorithm is designed to keep instantaneous variations small enough not to trigger classic threshold alarms, it cannot achieve perfect camouflage. The sequence of innovations loses its zero-mean

nature and begins to manifest a persistent bias, vectorially oriented in the direction of the deviation forced by the attack.

Since this single instantaneous alteration is too slight to arouse suspicion, it becomes crucial to monitor its cumulative trend over time using sequential statistical algorithms, such as the CUSUM (Cumulative Sum) test. This tool is not limited to evaluating the innovation of a single epoch but performs a continuous mathematical integration of it. While normal white noise tends to statistically cancel out over the long term when summed, the small but consistent directional bias introduced by spoofing accumulates monotonically. As soon as the sum of these consistent deviations exceeds a preset alert threshold, the algorithm identifies the vectorial anomaly and unmasks the intrusion, allowing the system to isolate the falsification before the kinematic deviation of the vehicle becomes physically irreversible.

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